

Quantitative Structural MRI Tissue Composition and FLAIR Signal Differences in Focal Cortical Dysplasia Using Interpretable Feature-Based Classification

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Abstract

Focal cortical dysplasia (FCD) is widely recognized as one of the most common structural causes of drug-resistant epilepsy and it is frequently associated with abnormalities in cortical organization that can be difficult to identify using visual inspection alone. Because these abnormalities are often subtle at the level of conventional anatomical MRI, I believe that quantitative structural measurements derived from multimodal MRI may help provide a useful complementary strategy for detecting some of the group-level differences between individuals who have FCD and healthy control subjects.

In this investigation, I analyzed multimodal structural MRI features that were derived from T1-weighted tissue segmentation together with intensity-based structural measurements extracted from FLAIR imaging in a balanced dataset of 170 subjects, which included 85 FCD subjects and 85 controls, all obtained from the OpenNeuro focal cortical dysplasia dataset. Whole-brain gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) tissue fractions were computed using SPM segmentation, while the global FLAIR intensity statistics and asymmetry-based measurements were all extracted after registration into subject-native anatomical space. Logistic regression models in combination with stratified five-fold cross-validation were then used to evaluate whether these interpretable structural measurements were able to support classification between different groups.

When looking across segmentation-derived measurements as a whole, I found that subjects with FCD appeared to show increased gray matter fraction together with reduced cerebrospinal fluid fraction relative to controls. At the same time, lesion regions themselves seemed to demonstrate consistently elevated FLAIR intensity relative to surrounding tissue within the same subjects. Classification using global structural features was able to produce cross-validated ROC-AUC values near 0.71, which I believe reflects the presence of moderate but interpretable discriminative structural signal across this dataset. Taken together, these observations help to support the idea that segmentation-derived tissue composition, when combined with lesion-level FLAIR contrast, can provide complementary quantitative structural markers of focal cortical dysplasia.

1 Introduction

Focal cortical dysplasia (FCD) is one of the most common structural causes of drug-resistant epilepsy and is frequently associated with abnormalities in cortical organization that can remain difficult to detect using conventional MRI inspection alone. When looking carefully at structural MRI in clinical practice, many FCD lesions appear subtle rather than overtly visible, which could suggest that quantitative structural measurements may help to provide additional information beyond the visual inspection alone.

In this study, I examined whether segmentation-derived tissue composition measurements from T1-weighted MRI together with intensity-derived structural measurements from FLAIR imaging could help distinguish individuals with focal cortical dysplasia from healthy control subjects within a balanced multimodal MRI dataset. In particular, I focused on whether whole-brain gray matter, white matter, and cerebrospinal fluid tissue fractions happened to differ systematically between groups and whether the lesion-level FLAIR signal contrast is able to provide additional discriminative information beyond the global tissue-composition measurements alone.

Because segmentation-derived tissue fractions summarize how structural tissue classes are distributed across the brain as a whole, I believe that they provide a useful starting point for evaluating whether or not global structural composition differs between different groups. At the same time, prior observations of lesion-associated hyperintensity patterns on FLAIR imaging would suggest that lesion-level contrast measurements might be able to capture localized structural signal changes that may not be fully reflected in the global tissue-composition summaries. Together, I believe that these complementary measurements provide an interpretable multimodal structural framework for characterizing focal cortical dysplasia across subjects within this dataset.

2 Methods

2.1 Dataset

Structural MRI data were obtained all from the OpenNeuro focal cortical dysplasia dataset (titled ds004199), which includes multimodal MRI scans from individuals with focal cortical dysplasia together with matched healthy control subjects. When organizing the dataset for this analysis, I confirmed that the sample included the 170 total subjects comprised of 85 FCD participants and 85 healthy controls, each with both T1-weighted MRI and FLAIR imaging available for multimodal structural comparison. Lesion region-of-interest (ROI) masks were available for FCD subjects and were used during lesion-level signal characterization.

Because the dataset followed a BIDS directory structure, subject-level imaging files could be identified systematically and then organized into a derivatives workspace prior to preprocessing. This definitely helped to ensure that the preprocessing steps remained consistent across subjects throughout the analysis pipeline.

2.2 Preprocessing

Structural preprocessing was all performed using MATLAB together with SPM segmentation and registration procedures.

FLAIR images were first coregistered into each subject’s native T1-weighted anatomical space. Because lesion the ROI masks needed to remain spatially aligned with the corresponding FLAIR images after transformation, some identical registration parameters were applied to the lesion masks during reslicing. Registration quality was then verified by comparing image dimensions, affine matrices, and the nonzero voxel counts within ROI masks.

Following the registration, T1-weighted images were segmented into gray matter (c1), white matter (c2), and cerebrospinal fluid (c3) tissue probability maps using SPM segmentation. These segmentation outputs then provided the basis for extracting whole-brain tissue-composition measurements used during feature construction.

2.3 Feature extraction

After the segmentation and the registration were completed, the structural features were extracted from both the segmentation-derived tissue probability maps and the registered FLAIR images.

Segmentation-derived features included whole-brain gray matter, white matter, and cerebrospinal fluid fractions together with related probability summary measurements. Because these tissue fractions describe how structural tissue classes are distributed across the brain as a whole, I believe that they provide a compact representation of the global structural composition differences between subjects.

Global FLAIR features included mean signal intensity, standard deviation, percentile-based intensity measurements, and peak axial slice intensity values. When looking across prior studies of focal cortical dysplasia, lesion-associated hyperintensity patterns on FLAIR imaging appear frequently, which suggests that summary intensity measurements capturing the upper range of signal variation may be able to reflect localized structural abnormalities.

Additionally, left-right asymmetry measurements were also computed for selected tissue-composition and FLAIR-derived features. Across subjects, these asymmetry measurements provided an additional representation of lateralized structural imbalance that may be associated with focal cortical dysplasia.

Lesion-level FLAIR contrast measurements were computed for subjects with available lesion masks by comparing mean signal intensity inside lesion regions with surrounding tissue regions within the same subjects.

2.4 Classification framework

Machine learning analyses were performed using logistic regression with stratified five-fold cross-validation.

Variables directly encoding lesion mask availability or lesion size were all removed prior to model training in order to avoid that sort of information leakage. Classification performance was evaluated using accuracy together with receiver operating characteristic area under the curve (ROC-AUC).

To better understand how the different feature groups ended up contributing to the classification performance, models were evaluated using segmentation-derived features alone and segmentation-plus-FLAIR feature sets.

3 Results

3.1 Whole-brain tissue composition differences

When comparing the segmentation-derived tissue fractions between all of the groups, I found that subjects with focal cortical dysplasia appeared to show increased gray matter fraction together with reduced cerebrospinal fluid fraction relative to healthy control subjects. Because tissue-fraction measurements summarize how structural tissue classes are distributed across the brain overall, this pattern would suggest that focal cortical dysplasia is associated with measurable differences in global structural composition.

White matter fraction measurements appeared more similar across groups, which suggests that the strongest segmentation-derived structural differences in this dataset were primarily associated with gray matter and cerebrospinal fluid measurements rather than with white matter structure.

3.2 Global FLAIR intensity differences

When examining the global FLAIR-derived intensity measurements across all the subjects, several summary intensity statistics appeared elevated in the focal cortical dysplasia group relative to controls. In particular, I found that the high-percentile intensity measurements together with peak axial slice intensity values differed between groups in a way that would suggest the presence of localized hyperintense signal structures distributed across affected brains.

Because percentile-based intensity measurements capture the upper range of signal variation within each subject’s FLAIR image, I believe these measurements reflect localized structural signal differences rather than uniform whole-brain intensity shifts.

3.3 Classification performance

Logistic regression models using segmentation-derived tissue-composition features alone achieved cross-validated ROC-AUC values near 0.71. Because these models relied only on interpretable whole-brain tissue-composition measurements, I think this result would suggest that segmentation-derived structural features already contain a measurable discriminative signal separating groups.

When global FLAIR intensity features were added to segmentation-derived measurements, classification performance changed only slightly. This observation would suggest that segmentation-derived tissue-composition features already captured much of the classification-relevant signal present at the whole-brain level within this dataset.

3.4 Lesion-level FLAIR contrast differences

When comparing lesion-level signal intensity within subjects with available lesion masks, mean FLAIR intensity inside lesion regions appeared substantially higher than the outside lesion regions within the same subjects ($p \approx 2 \times 10^{-19}$). Because this comparison examined within-subject signal differences

rather than between-group differences, I believe it provides direct structural evidence that the lesion regions consistently show elevated FLAIR signal relative to surrounding tissue.

3.5 Lesion tissue composition overlap

Segmentation-derived tissue probability overlap measurements showed that lesion voxels were located primarily within gray matter probability regions, with smaller contributions from white matter and minimal overlap with cerebrospinal fluid probability regions. This distribution is consistent with the expectation that focal cortical dysplasia primarily affects cortical gray matter structure.

4 Conclusion

Across all of the segmentation-derived tissue-composition measurements as a whole, I found that subjects with focal cortical dysplasia appeared to show increased gray matter fraction together with reduced cerebrospinal fluid fraction relative to the healthy controls. Because tissue-fraction measurements summarize how the structural tissue classes are distributed across the brain overall, this pattern would perhaps suggest that focal cortical dysplasia is associated with measurable global structural composition differences detectable using segmentation-based approaches.

At the same time, lesion-level analyses happened to show that FLAIR intensity values within lesion regions remained consistently elevated relative to surrounding tissue within the same subjects. I believe these lesion-specific signal differences provide a direct structural marker of localized abnormal signal intensity patterns characteristic of focal cortical dysplasia.

When evaluating classification performance using interpretable feature-based logistic regression models, segmentation-derived tissue-composition measurements alone supported moderate classification performance with ROC-AUC values near 0.71. Adding global FLAIR-derived intensity measurements produced only modest improvement, which would suggest that segmentation-derived measurements were able to capture much of the classification-relevant structural signal that is present at the whole-brain level.

Taken together, these results support the interpretation that segmentation-derived tissue fractions together with lesion-level FLAIR contrast are able to provide a useful interpretable multimodal structural framework for characterizing focal cortical dysplasia across subjects within this dataset.

References

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Figures

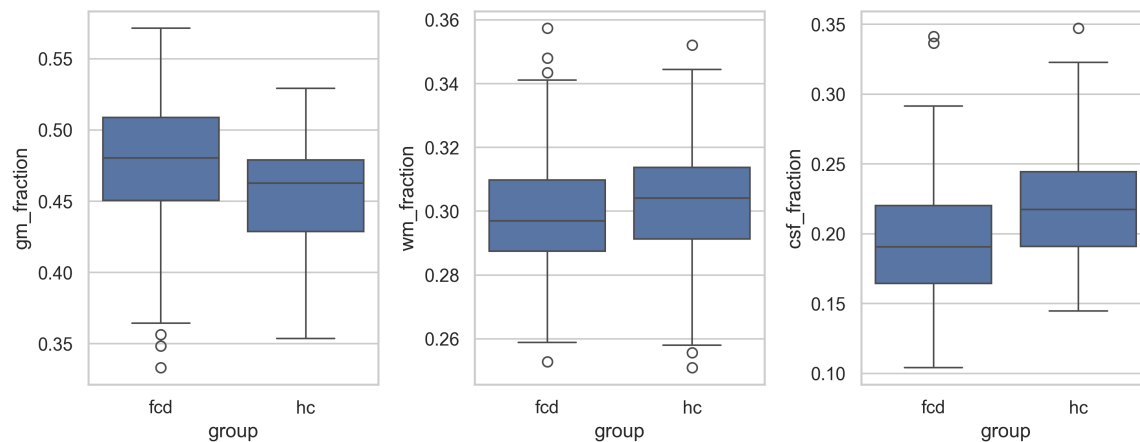


Figure 1: Group comparison of segmentation-derived tissue fractions across subjects. FCD subjects show increased gray matter fraction and reduced cerebrospinal fluid fraction relative to healthy control subjects.

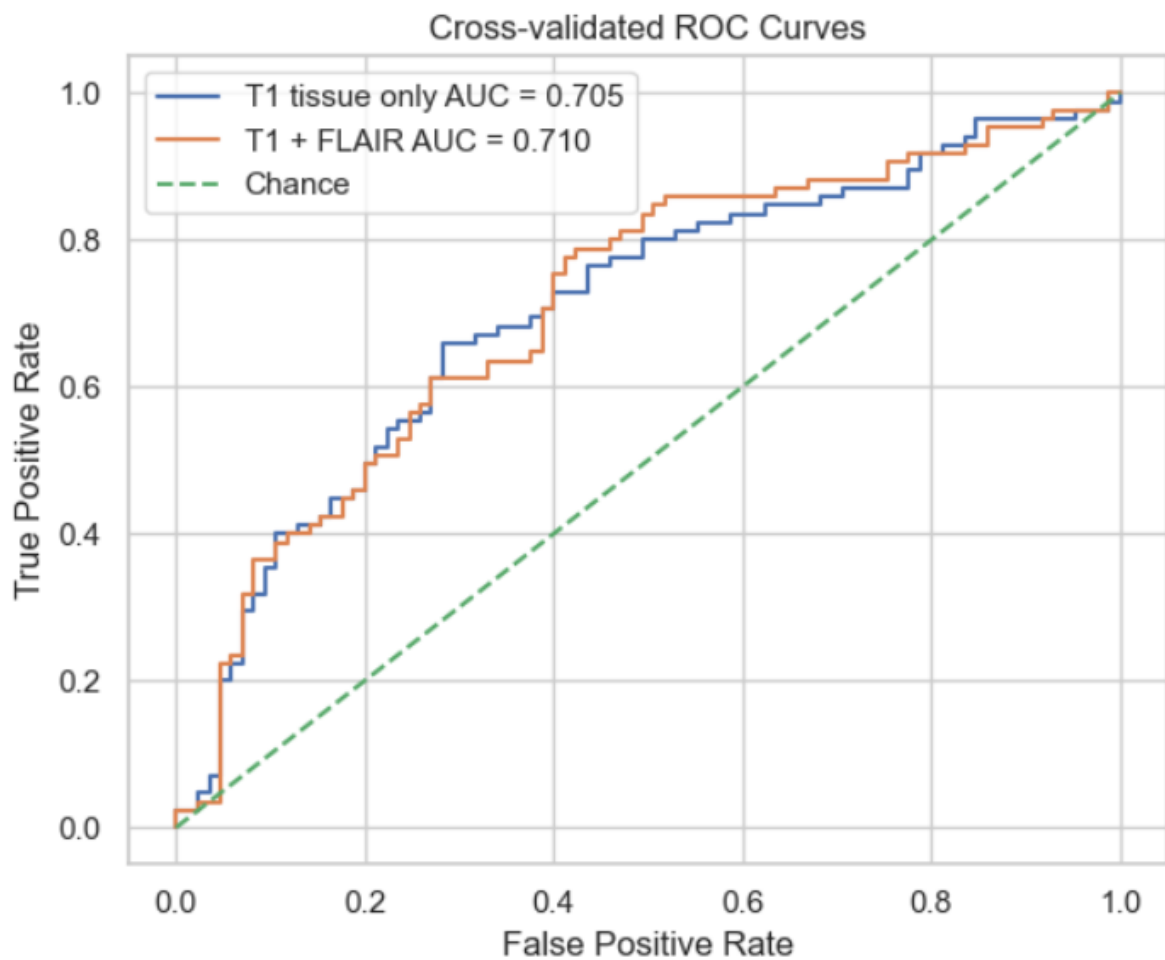


Figure 2: Cross-validated ROC curves for logistic regression classification using segmentation-derived tissue composition together with global FLAIR-derived intensity features. Overall classification performance produced ROC-AUC values near 0.71.

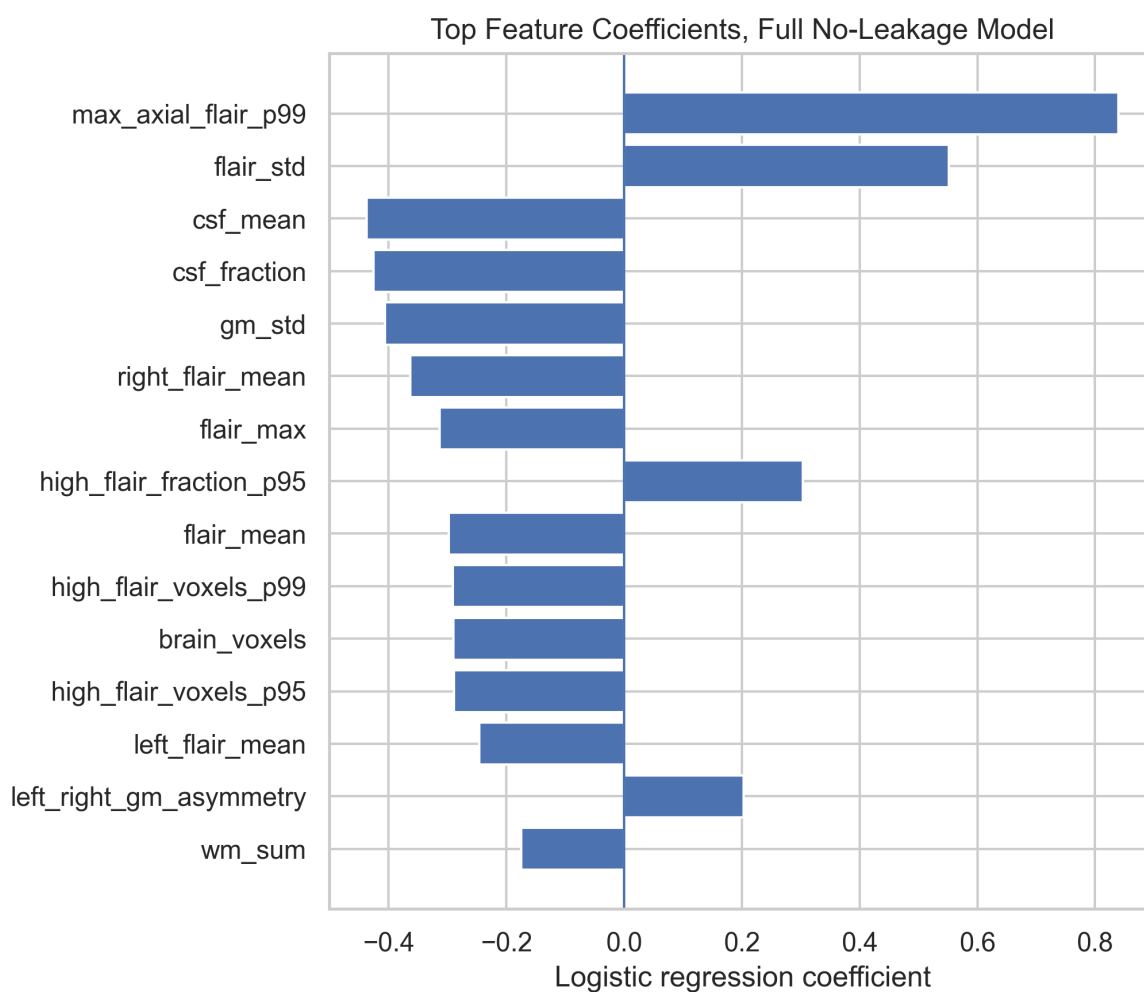


Figure 3: Logistic regression feature coefficients from the interpretable structural feature model. Gray matter fraction contributed the strongest classification signal among segmentation-derived tissue-composition features.

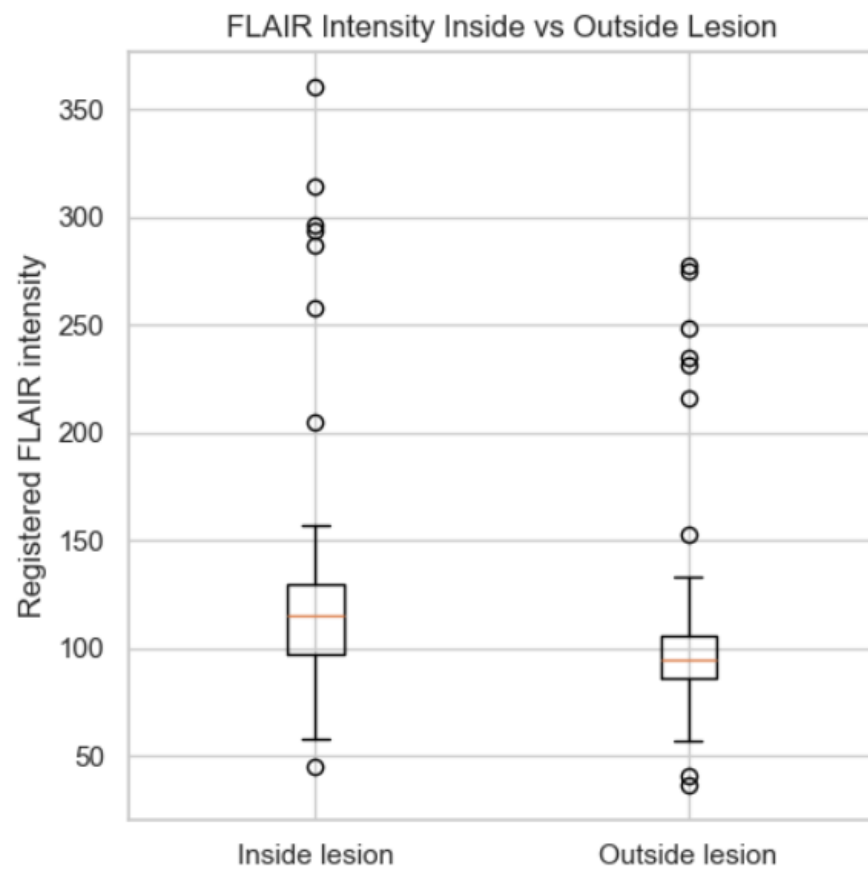


Figure 4: Comparison of FLAIR intensity inside lesion masks versus surrounding tissue within the same subjects. Lesion regions show elevated FLAIR signal intensity relative to adjacent anatomical regions.

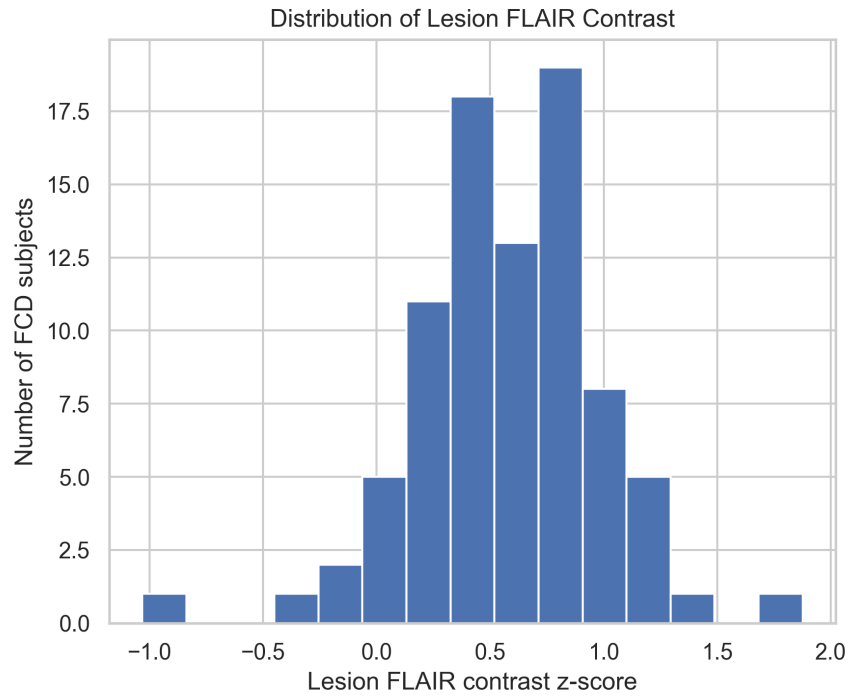


Figure 5: Distribution of lesion-level FLAIR contrast differences across FCD subjects. Elevated lesion signal intensity relative to surrounding tissue provides a localized structural marker of focal cortical dysplasia.

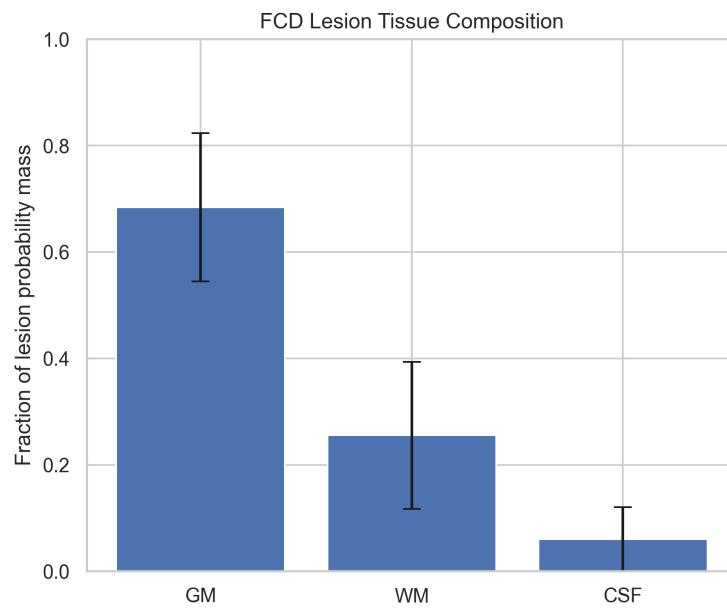


Figure 6: Average segmentation-derived tissue probability composition within FCD lesion masks. Lesion regions were composed primarily of gray matter probability mass with smaller white matter contribution and minimal cerebrospinal fluid overlap.